

Jonathan Kwan

Full-Stack Software Engineer

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in [LinkedIn](#)

🐙 [GitHub](#)

🌐 [Portfolio Website](#)

TECHNICAL SKILLS

Programming Languages | Python, JavaScript, SQL, HTML, CSS

Frameworks & Tools | FastAPI, React, SQLAlchemy, Pydantic, Vite, Docker (with Docker Compose), Git, Render

Databases & APIs | PostgreSQL, RESTful API design, OpenAI API

AI/ML Exploration | Google AI Essentials, OpenAI API



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RELEVANT PROJECTS

WasteNot | Backend Engineer

Jun '25

<https://www.wastenotkitchen.com>

<https://github.com/kimmiwong/wastenot>

Stack: FastAPI · SQLAlchemy · PostgreSQL · Docker · Spoonacular API ·

Google Maps API · React · Vite · React Router · Git · Render

This app helps users track food items by expiration date, generate recipes based on selected ingredients, and reduce waste through smart composting and meal planning.

- **Built FastAPI + SQLAlchemy backend** to manage household data, automate expirations, and trigger real-time alerts
- **Integrated the Spoonacular API** to generate personalized recipe recommendations based on users' selected ingredients, reducing food waste
- **Implemented secure authentication** using **session cookies** and **bcrypt password hashing** to protect user accounts and sessions
- **Deployed full-stack app** to Render, managing **React frontend**, **FastAPI backend**, and **PostgreSQL database** services
- **Configured environment variables and API keys** with **python-dotenv** for secure local and cloud deployment

Expense Tracker Lite | Full Stack Engineer

May '25

<https://github.com/jkwan14/expense-tracker-lite>

Stack: FastAPI · SQLAlchemy · PostgreSQL · Docker · OpenAI API · React ·

React Router · CSS

This app helps users track and categorize daily expenses using AI to automatically classify transactions for smarter budgeting.

- **Engineered AI-powered categorization** using **GPT-3.5** with **NLP** and **Named Entity Recognition (NER)** for improved accuracy
- **Designed custom prompt logic with fallback handling**, tested on 100+ varied transaction descriptions for consistent results
- **Developed RESTful API endpoints in FastAPI** for transactions and categories, supporting creation, retrieval, filtering, and deletion
- **Constructed modular backend architecture** with **SQLAlchemy models** and **Pydantic schemas** for clean data validation
- **Secured API keys and database configurations** with **python-dotenv** to minimize credential exposure risk

PROFESSIONAL EXPERIENCE

CHRISTINE M. REARDON-DAVIS, DMD | Website Administrator

Feb '24 - Feb '25

- **Redesigned and maintained mobile-friendly website**, improving UX and **increasing** online appointment requests
- Created **secure online forms** for patient intake and feedback, **improving HIPAA compliance** and reducing the need for paper-based processing
- Cooperated with the dental team to update bios, FAQs, and services, ensuring accuracy and supporting **20+ monthly online inquiries**

CHRISTINE M. REARDON-DAVIS, DMD | Lead Assistant

Apr '04 - Feb '24

- **Supervised and trained** a team of **4+ assistants**, ensuring proper sterilization, pediatric techniques, and **efficient daily workflows**
- Guided children with **anxiety or special needs** using tailored strategies, resulting in fewer behavioral disruptions and more **efficient procedures**
- **Coordinated** front desk and dental team to manage patient flow, **minimizing delays** and helping maintain a calm, child-friendly environment

Generations Mental Health | Office Manager

Aug '12 - Aug '23

- **Improved operations by implementing an EHR system**, improving scheduling, billing, and documentation for **30+ daily patients**
- **Managed billing, insurance claims**, and follow-ups for **100+ patients**, ensuring accuracy and consistent revenue
- **Handled 20+ daily patient inquiries** with empathy and discretion as the primary contact in a confidential mental health setting

EDUCATION

Hack Reactor | Full Stack Software Engineering Certificate - Software Engineering Immersive Program

Jun '25

- Full-Stack Web Development (JavaScript, Python, React)
- Backend Engineering (SQL, PostgreSQL, RESTful APIs, FastAPI)
- Frontend Design & UX (HTML/CSS, DOM manipulation, state management)
- Built a full-stack budgeting app with OpenAI integration for auto-categorization of expenses

George Mason University | Bachelor of Science - Psychology

Jan '11

- Cognitive Psychology (applied understanding of human behavior and decision-making)
- Research Methods & Statistics (hypothesis testing, data analysis)
- Developmental Psychology (understanding user needs across life stages—valuable for UX)
- Dean's List